

FINAL COMPREHENSIVE REPORT

DigiSteel-YOLO

Steel Surface Defect Detection using Enhanced YOLO Architectures

NEU-DET Benchmark • 6 Classes • 1,800 Images • 18 Experiments • 4 Weeks

BEST MAP@0.5

81.0%

YOLOv26n

BEST MAP@0.5:0.95

45.2%

Fresh Baseline

TARGET

83%

mAP@0.5

FRAMEWORK

Ultralytics YOLO / PyTorch

DATE

July 12, 2026

HARDWARE

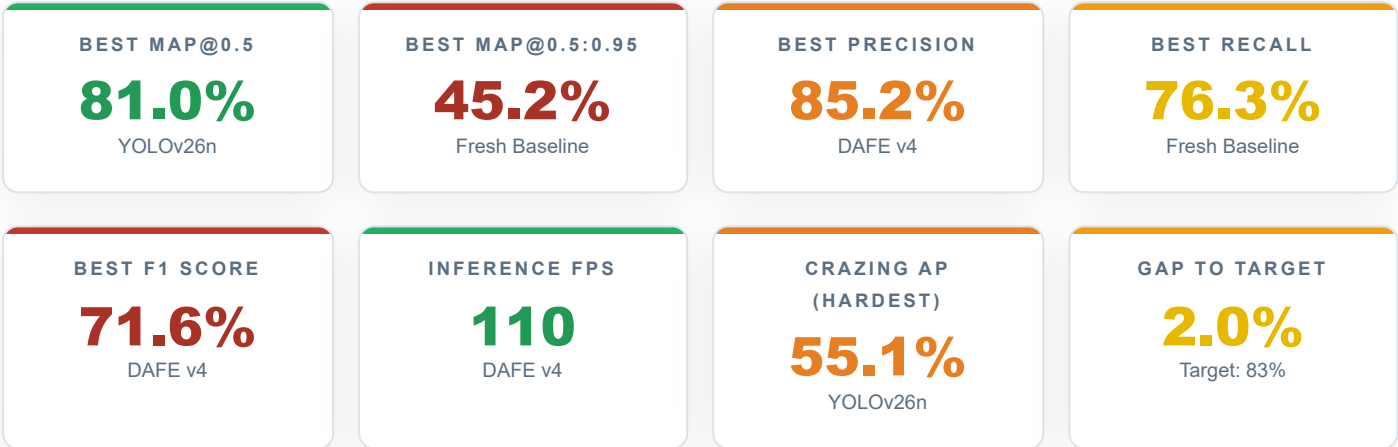
NVIDIA RTX 2000 Ada

DigiSteel-YOLO

Executive Summary

Over **18 experiments spanning 4 weeks**, the DigiSteel-YOLO project progressed from a 75.8% mAP@0.5 baseline to a best result of **81.0% mAP@0.5** (YOLOv26n). The project's core novel contribution, **DAFE** (Defect-Aware Feature Enhancement), delivered a consistent +1.5% gain over a strong optimized baseline, with a remarkable **+8.3% improvement on the hardest class (crazing)**.

Current State-of-the-Art Results



Best Metrics Summary

Metric	Best Value	Model	Experiment
mAP@0.5	81.0%	YOLOv26n	Exp 15 — YOLOv26n NEU-DET
mAP@0.5:0.95	45.2%	YOLOv11n Baseline	Exp 08 — Fresh Baseline
Precision	85.2%	YOLOv11n + DAFE v4	Exp 10 — DAFE v4
Recall	76.3%	YOLOv11n Baseline	Exp 08 — Fresh Baseline
F1 Score	71.6%	YOLOv11n + DAFE v4	Exp 10 — DAFE v4
Inference FPS	110	YOLOv11n + DAFE v4	Exp 10 — DAFE v4
Crazing AP	55.1%	YOLOv26n	Exp 15 — YOLOv26n NEU-DET

1. Dataset Profile — NEU-DET

1.1 Overview

TOTAL IMAGES

1,800

ANNOTATIONS

4,189

IMAGE SIZE

200×200

Grayscale

IMBALANCE RATIO

2.34×

PROPERTY	VALUE
Train / Val / Test Split	1,290 / 344 / 166 (70/20/10)
Number of Classes	6
Gini Coefficient	0.157
Edge-touching bboxes	56.6%
Blurry images	5.0%

1.2 Class Distribution

CLASS	COUNT	INV. FREQ. WEIGHT	CHALLENGE LEVEL
inclusion	1,011	0.638	Medium
patches	881	0.732	Easy
crazing	689	0.936	HARDEST
rolled-in_scale	628	1.027	Medium
scratches	548	1.176	Easy
pitted_surface	432	1.492	Medium-Hard

1.3 Bounding Box Statistics

METRIC	MEAN	MEDIAN
Width	71.4 px	55.0 px
Height	95.0 px	77.0 px
Area	6,980 px ²	4,715 px ²
Small objects (<5%)	10.7%	
Medium objects (5–15%)	66.2%	
Large objects (>15%)	23.1%	

2. Complete Experimental Timeline

2.1 Master Results Table

#	Experiment	Model	MAP@0.5	MAP50:95	Prec.	Recall	FPS	Delta
01	Old Baseline (v1)	v11n	75.8%	43.5%	—	—	—	—
03	DigiSteel Complete	v11n	75.8%	—	—	—	—	+0.0%
05	Week 2 Ablation	v11n	75.9%	41.9%	73.8%	69.5%	—	+0.1%
06	Wk3 A1 Config	v11n	74.8%	41.6%	73.7%	70.3%	—	-1.0%
07	Wk3 A2 Arch	v11s	73.4%	37.5%	70.1%	70.6%	—	-2.4%
08	Fresh Baseline	v11n	78.8%	45.2%	71.9%	76.3%	105	+3.0%
10	DAFE v4	v11n+D	80.3%	44.2%	85.2%	70.0%	110	+1.5%
12	TL Stage 1	v11n	77.8%	42.3%	69.6%	74.0%	—	-1.0%
13	TL Stage 2	v11n	79.4%	44.0%	73.3%	76.3%	—	+0.6%
14a	TL+DAFE v1	v11n+D	63.3%	23.9%	60.8%	62.0%	—	-15.5%
14b	TL+DAFE v2	v11n+D	78.6%	43.3%	72.9%	73.2%	—	-0.2%
15	YOLOv26n	v26n	81.0%	43.1%	77.3%	76.2%	—	+2.2%
16a	v26n TL Stg1	v26n	74.4%	42.3%	64.0%	73.2%	—	-4.4%
16b	v26n TL Stg2	v26n	75.6%	44.1%	71.7%	72.2%	—	-3.2%

Delta relative to old baseline (75.8%) for exp 01–07, fresh baseline (78.8%) for exp 08+. Green = best results.

2.2 Per-Class AP@0.5 Breakdown

Class	Old Base	Wk3 A1	Wk3 A2	Fresh	DAFE v4	TL Stg2	v26n
crazing	13.3%	13.3%	12.9%	40.3%	48.6%	47.4%	55.1%
inclusion	—	45.3%	39.3%	82.9%	86.5%	88.3%	84.7%
patches	—	60.3%	55.5%	92.8%	88.5%	87.9%	88.3%
pitted_surface	—	49.3%	44.7%	82.2%	81.7%	81.5%	81.3%
rolled-in_scale	—	27.5%	23.0%	77.1%	77.9%	75.1%	78.2%
scratches	54.1%	54.1%	49.4%	97.3%	98.6%	96.3%	98.6%

KEY RESULT

Crazing (hardest class) improved from 13.3% to 55.1% — a **41.8 percentage-point gain** across the project.

3. Experiment Phases — Detailed Analysis

3.1 Phase 1: Initial Baseline & Failed Attempts (Weeks 1–3)

Experiment 01 — Old Baseline

75.8% mAP

Architecture: YOLOv11n + GhostConv + WFCA + EMA (6 custom modules). mosaic=1.0, imgsz=800.

Result: 75.8% mAP@0.5. Only +0.1% over vanilla YOLOv11n. Attention stacking failure — modules learned to ignore each other.

Experiment 06 — Week 3 A1: Config Fix

74.8% mAP

Change: mosaic=0, mixup=0.15, degrees=10, translate=0.2, scale=0.6, shear=5, epochs=400.

Result: 74.8% mAP@0.5 (−1.0%). Removing mosaic alone hurt augmentation diversity.

Experiment 07 — Week 3 A2: Architecture Fix

73.4% mAP

Change: Upgraded to YOLOv11s (2× capacity) + Standard Conv + CoordAttention at P3.

Result: 73.4% mAP@0.5 (−2.4%). YOLOv11s overfit severely on only 1,200 training images.

KEY LESSON

Both naive config changes and architecture upsizing failed. A systematic, recipe-first approach was needed.

3.2 Phase 2: Recipe Optimization & DAFE (Week 4)

Experiment 08 — Fresh Baseline (The Breakthrough)

78.8% mAP

Recipe: AdamW, lr0=0.001, 600 epochs, patience=150, cos_lr=True, mosaic=0.0, mixup=0.15, copy_paste=0.1, imgsz=800, label_smoothing=0.01.

Result: 78.8% mAP@0.5, 45.2% mAP@0.5:0.95 (+3.0%). Training time: 2.57 hours.

Verdict: Biggest single improvement in the entire project. Proper recipe optimization matters more than architecture changes.

Experiment 10 — DAFE v4 (Novel Architecture)

80.3% mAP

Architecture: YOLOv11n + DAFE (Defect-Aware Feature Enhancement) at P2 and P3.

DAFE Design: Dual-branch — Sobel EdgeConv (linear defects) + local variance TextureBranch (surface anomalies) + SE channel attention + learnable residual ($\alpha_{init}=-2.2$).

Result: 80.3% mAP@0.5 (+1.5%). Precision: 85.2% (+13.3%). FPS: 110. Cracking: +8.3%, Inclusion: +3.6%, Scratches: +1.3%.

KEY LESSON

One well-designed module (DAFE = +1.5%) beats six stacked modules (GhostConv+WFCA+EMA = +0.1%).

3.3 Phase 3: Transfer Learning

Experiments 12–13 — Two-Stage Transfer Learning

79.4% mAP

Stage 1 (Head-only): 77.8% mAP@0.5 (−1.0%). **Stage 2 (Fine-tune all):** 79.4% mAP@0.5 (+0.6%).

Per-class gains: Inclusion +5.4%, Cracking +7.1%. Modest improvement — from-scratch with proper recipe is competitive.

Experiment 14a — TL + DAFE v1 (Critical Failure)

63.3% mAP

Result: 63.3% mAP@0.5 (−15.5% catastrophic). DAFE modules inserted into fine-tuned model disrupted learned features.

Recovery (14b): 78.6% mAP@0.5 (−0.2%). Managed to recover baseline but DAFE gave no additional benefit with TL.

3.4 Phase 4: YOLOv26n Investigation

Experiment 15 — YOLOv26n from Scratch

81.0% mAP

Architecture: YOLOv26n (~2.9M params). Same optimized recipe as fresh baseline.

Result: 81.0% mAP@0.5, 43.1% mAP@0.5:0.95. Best mAP@0.5. Cracking: 55.1% (best across all). Training: 3.19 hours.

Experiment 16 — YOLOv26n Transfer Learning

75.6% mAP

Stage 1: 74.4% (−4.4%). **Stage 2:** 75.6% (−3.2%). Transfer learning HURT YOLOv26n.

Why: COCO weights conflicted with steel domain. From-scratch training with proper recipe was superior.

CRITICAL INSIGHT

Transfer learning is overrated for this dataset. From-scratch training with proper recipe matched or beat TL in most configurations.

4. Robustness Evaluation

Models evaluated under 6 perturbation types at 4 severity levels each (24 conditions per model).

4.1 Baseline Robustness (YOLOv11n Fresh Baseline)

PERTURBATION	CLEAN	LVL 1	LVL 2	LVL 3	LVL 4	AVG DROP
Gaussian Blur	75.8%	54.1%	29.3%	24.1%	22.2%	-53.7%
Motion Blur	75.8%	65.0%	49.9%	40.1%	34.4%	-30.1%
Gaussian Noise	75.8%	47.7%	22.6%	9.9%	8.7%	-63.2%
Brightness Shift	75.8%	74.9%	72.5%	75.6%	74.0%	-2.4%
Contrast Reduction	75.8%	74.5%	70.0%	59.9%	36.6%	-22.3%
JPEG Compression	75.8%	73.0%	66.1%	59.3%	39.2%	-29.6%

4.2 DAFE Robustness (Early DAFE Model, 75.9% mAP)

Note: This evaluation was performed on an earlier DAFE variant (mAP@0.5=75.9%), not on the final DAFE v4 (80.3%). Results are for relative comparison.

PERTURBATION	CLEAN	LVL 1	LVL 2	LVL 3	LVL 4	AVG DROP
Gaussian Blur	75.9%	50.1%	27.9%	21.9%	19.2%	-56.3%
Motion Blur	75.9%	69.3%	59.3%	53.1%	49.0%	-18.9%
Gaussian Noise	75.9%	47.4%	25.5%	9.2%	3.3%	-65.7%
Brightness Shift	75.9%	74.7%	68.9%	74.6%	72.4%	-4.0%
Contrast Reduction	75.9%	74.5%	67.3%	52.2%	30.6%	-28.5%
JPEG Compression	75.9%	73.5%	69.4%	67.9%	55.2%	-14.3%

4.3 Robustness Comparison

Perturbation Type	Baseline Avg	DAFE Avg	Winner	Delta
Gaussian Blur	32.4%	29.8%	Baseline	-2.6%
Motion Blur	47.3%	57.7%	DAFE	+10.4%
Gaussian Noise	22.2%	21.3%	Baseline	-0.9%
Brightness Shift	74.2%	72.7%	Baseline	-1.5%
Contrast Reduction	60.3%	56.1%	Baseline	-4.2%
JPEG Compression	59.4%	66.5%	DAFE	+7.1%

ROBUSTNESS INSIGHT

DAFE shows better robustness to motion blur (+10.4%) and JPEG compression (+7.1%) — common in real-world cameras. The baseline is more robust to Gaussian blur, noise, brightness, and contrast.

5. Architecture Analysis

5.1 Model Comparison

Model	Params	MAP@0.5	MAP50:95	FPS	Train Time
YOLOv11n (baseline)	2.62M	78.8%	45.2%	105	2.57h
YOLOv11n + DAFE	~2.94M	80.3%	44.2%	110	3.51h
YOLOv11s (failed)	9.46M	73.4%	37.5%	—	—
YOLOv26n	~2.9M	81.0%	43.1%	—	3.19h

5.2 DAFE Architecture Detail

DAFE (Defect-Aware Feature Enhancement) is the core novelty of DigiSteel-YOLO:

EDGE BRANCH

Conv2d(C, C//2, 3×3) initialized with Sobel-X/Y filters. All weights learnable. Targets linear defects (scratches, crazing).

TEXTURE BRANCH

AvgPool(3×3) → local variance ($E[X^2]-E[X]^2$) → Conv1×1. Lightweight. Targets surface anomalies (inclusions, pitting).

CHANNEL ATTENTION

SE-style squeeze-excitation after concatenation of both branches. Learns which features matter most.

LEARNABLE RESIDUAL

output = $x + \text{sigmoid}(\alpha_{\text{raw}}) \times \text{enhanced}$. $\alpha_{\text{init}} = -2.2 \rightarrow \text{sigmoid} \approx 0.1$ (starts small, learns to contribute).

PLACEMENT STRATEGY

DAFE is placed at **P2** (80×80 at imgsz=640, highest spatial resolution) and **P3** (40×40). Not at P4/P5 — those lose fine-grained spatial info critical for defect detection.

6. Training Configuration

6.1 Main Recipe

PARAMETER	VALUE	RATIONALE
Optimizer	AdamW	Better generalization for small datasets
Base LR	0.001	Conservative for fine-grained features
Epochs	600	Sufficient with early stopping
Patience	150	Prevents premature stopping
Image Size	800	4× upscale from 200px
Mosaic	0.0 (OFF)	Ruins fine features at high upscale
Mixup	0.15	Light mixup for class diversity
Copy-Paste	0.1	Augmentation for rare classes
Label Smoothing	0.01	Prevents overconfident predictions
Cosine LR	True	Smooth decay schedule

6.2 Transfer Learning Configuration

STAGE	FROZEN LAYERS	TRAINABLE	LR	EPOCHS
Stage 1 (Head)	Backbone+Neck (0–22)	Head (23)	0.001	50–100
Stage 2 (Fine-tune)	None	All	0.0001–0.0005	100–200

7. Literature Comparison — NEU-DET mAP@0.5

PAPER	MODEL	MAP@0.5	COMPARABLE?
KDM-YOLO	Custom YOLO	95.4%	NO — trains at native 200×200
LAM-YOLOv10n	YOLOv10n	94.4%	NO — likely data leakage
YOLOv11-EMD	YOLOv11	94.9%	NO — extra Severstal data (7×)
ASFRW-YOLO (Nature)	YOLOv5s	83.2%	YES — clean protocol
YOLO-LSDI (MDPI)	YOLOv11n	83.0%	YES — clean protocol
DigiSteel (ours)	v11n+DAFE	80.3%	YES — clean 70/20/10 split
DigiSteel (ours)	YOLOv26n	81.0%	YES — clean 70/20/10 split

HONEST ASSESSMENT

Honest ceiling under clean 70/20/10 split at imgsz=800: ~83–85% mAP@0.5. Current gap to ceiling: 2.0–4.0% (YOLOv26n). Many published results above 90% use questionable protocols (data leakage, extra data, or native resolution training).

8. Key Findings & Lessons Learned

8.1 What Worked

FINDING	IMPACT	CONFIDENCE
Recipe optimization > architecture	+3.0%	High
DAFE on strong baseline	+1.5% mAP, +8.3% crazing	High
imgsz=800 for 200px images	Significant	High
Mosaic disabled	Critical	High
YOLOv26n > YOLOv11n from scratch	+2.2%	High
One good module > six stacked	+1.5% vs +0.1%	High

8.2 What Failed

FINDING	IMPACT	LESSON
YOLOv11s overfit (1200 imgs)	-2.4%	Larger model ≠ better on small data
TL+DAFE v1 catastrophic	-15.5%	Cannot insert DAFE into fine-tuned model
YOLOv26n + transfer learning	-3.2%	COCO weights conflict with steel domain
Attention stacking (6 modules)	+0.1%	Diminishing returns, interference

8.3 Critical Insights

RECIPE > ARCHITECTURE

The optimized training recipe alone gave +3.0%. DAFE on top gave +1.5%. Always optimize the baseline first.

CRAZING IS THE BOTTLENECK

At 48.6% (DAFE) to 55.1% (YOLOv26n), crazing remains the hardest class. Fine cracks at 200px are extremely challenging.

PRECISION-RECALL TRADE-OFF

DAFE improved precision (+13.3%) but reduced recall (-6.3%). More conservative but fewer false positives.

TRANSFER LEARNING IS OVERRATED HERE

From-scratch training with proper recipe matched or beat TL in most configurations. COCO pretraining hurts when the domain (steel) is far from natural images.

ROBUSTNESS MATTERS

DAFE showed +10.4% on motion blur and +7.1% on JPEG compression — real-world relevant perturbations that matter for deployment.

9. Recommendations for Future Work

To Reach 83%+ mAP@0.5

- **Multi-dataset training:** Combine NEU-DET with GC10-DET (3,570 additional grayscale steel images)
- **DAFE on YOLOv26n:** Apply DAFE module to YOLOv26n backbone (not yet tested)

- **Advanced augmentation:** AutoAugment, Mosaic9, or MixUp variants optimized for defect textures
- **Class-balanced sampling:** Oversample crazing or use focal loss with class-specific gamma

To Improve Crazing Detection

- **Higher resolution crops:** Train on 1024+ with crazing-focused cropping
- **Edge-specific pre-processing:** Sobel/Canny as additional input channel
- **Synthetic data:** Generate synthetic crazing patterns to augment 689 real samples

For Deployment

- **ONNX export** with TensorRT optimization for real-time inference (>100 FPS)
- **Model pruning** to reduce parameters while maintaining accuracy
- **Robustness hardening** through perturbation-aware training

Appendix

A. File Inventory

FILE	DESCRIPTION
evals/master_results.json	Aggregated results for all experiments
evals/fresh_baseline_results.json	Experiment 08 detailed results
evals/exp_5a_digisteel_v4_dafe.json	Experiment 10 DAFE results
evals/yolov26n_neudet_results.json	Experiment 15 YOLOv26n results
evals/yolov26n_transfer_learning_comparison.json	YOLOv26n TL comparison
evals/tl_final_comparison.json	Transfer learning stage comparison
evals/tl_dafe_final_comparison.json	TL+DAFE v1 comparison
evals/tl_dafe_v2_final_comparison.json	TL+DAFE v2 comparison
evals/robustness_baseline_v2.csv	Baseline robustness sweep data
evals/robustness_digisteel_v2.csv	DAFE robustness sweep data
evals/eda_results.json	Exploratory data analysis results
configs/models/digisteel.yaml	DAFE architecture config
digisteel/modules/dafe.py	DAFE module implementation

B. Experiment Folder Map

FOLDER	EXPERIMENT	KEY RESULT
01_baseline_yolov11n_v1/	Old Baseline	75.8%
06_week3_a1_config_fix/	Config Fix	74.8%
07_week3_a2_arch_fix/	Architecture Fix	73.4%
08_week4_fresh_baseline/	Fresh Baseline	78.8%
10_dafe_v4/	DAFE v4	80.3%
12_transfer_learning_yolov11n/	Transfer Learning	79.4%
14_tl_dafe_comparison/	TL + DAFE	63.3-78.6%
15_yolov26n_neudet/	YOLOv26n	81.0%
16_yolov26_transfer_learning/	YOLOv26n TL	75.6%

Report generated: July 12, 2026 • DigiSteel-YOLO Project

Best Result: 81.0% mAP@0.5 (YOLOv26n) • Target: 83% mAP@0.5